

# Nelson Cala-Moreno

DVM · MSc · PhD

Quantitative Genetics · Biostatistics · Genomics · Animal Science

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|---------------|--------------------------------------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------|
| Location      | Bucaramanga, Santander, Colombia                                               | ORCID   | 0000-0002-9439-7007                                                                                                 |
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## PROFESSIONAL PROFILE

Doctor of Veterinary Medicine and PhD in Animal Production and Health Sciences from the Universidad Nacional Autónoma de México (UNAM), with master's and doctoral training focused on genetics and biostatistics applied to animal sciences.

Experienced in university teaching, research and the supervision of academic work in genetics, molecular biology, quantitative genetics, genomics, biostatistics, experimental design and research methodology. Has participated in scientific projects on genetic characterisation, genomic analysis, estimation of genetic parameters, animal adaptation and health under tropical conditions.

Skilled in the analysis of complex biological data through mixed models, multivariate techniques, genome-wide association studies (GWAS) and advanced statistical tools. His academic work integrates the teaching of biological and genetic fundamentals with quantitative methodologies applied to veterinary research.

## ACADEMIC INDICATORS

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|---------------------------------------------------------------|-------------------------------------------------------|
| <b>56</b> citations (Google Scholar)                          | <b>PhD</b> — UNAM, Mexico                             |
| <b>h-index: 4 · i10-index: 1</b>                              | <b>MSc</b> — UNAM, Mexico                             |
| <b>20+</b> supervised theses and dissertations completed      | National and international peer-reviewed publications |
| Principal investigator and co-investigator on funded projects | Peer reviewer for scientific journals                 |

## EDUCATION

### PhD in Animal Production and Health Sciences

Universidad Nacional Autónoma de México (UNAM), Mexico City · 2021

- Research line: Animal Genetics and Biostatistics.
- CONACYT scholarship holder throughout the doctoral programme.
- Doctoral thesis: *Estimation of genotype-by-environment interaction effects for body weight and survival at harvest in two genetic lines of Pacific white shrimp* *Penaeus* (*Litopenaeus*) *vannamei*.
- Degree officially recognised in Colombia through Resolution No. 007447 of 5 May 2023, Ministry of National Education.

### MSc in Animal Production and Health Sciences

Universidad Nacional Autónoma de México (UNAM), Mexico City · 2014

- Research line: Animal Genetics and Biostatistics.
- CONACYT scholarship holder throughout the master's programme.
- Master's thesis: *Evolution of genetic variability for body weight at commercial age in a selected population of Pacific white shrimp* *Penaeus* (*Litopenaeus*) *vannamei*.
- Degree officially recognised in Colombia through Resolution No. 015387 of 25 August 2020, Ministry of National Education.

## Doctor of Veterinary Medicine (DVM)

Universidad Cooperativa de Colombia (UCC), Bucaramanga · 2005

- Undergraduate thesis: *Development of an epidemiological map of zoonotic diseases in Santander, Colombia.*

## COMPLEMENTARY TRAINING

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1. "Phenotypic Prediction Workshop". Webinar — University of Florida. 08/2015
2. "Design of Breeding Programs with Genomic Selection". University of New England. 02/2014
3. "Linear and Generalised Mixed Models using ASReml and Meta-analysis in the Biological Sciences". Faculty of Veterinary Medicine and Animal Science — Universidad Nacional Autónoma de México. 05/2011
4. "Basic Techniques in Molecular Biology". Faculty of Veterinary Medicine and Animal Science — Universidad Nacional Autónoma de México. 10/2010
5. "Hazardous Waste Management in the Department of Genetics and Biostatistics, FMVZ". Faculty of Veterinary Medicine and Animal Science — Universidad Nacional Autónoma de México. 10/2010
6. "An Introduction to Population Genetics Theory: 100 Years after Hardy and Weinberg". Faculty of Veterinary Medicine and Animal Science — Universidad Nacional Autónoma de México. 08/2010
7. "The R Software in the Basic Teaching of Statistical Methods". Faculty of Veterinary Medicine and Animal Science — Universidad Nacional Autónoma de México. 07/2010

## SCIENTIFIC OUTPUT

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### Published Theses

1. Cala-Moreno, N. (2019). *Estimación de efectos de interacción genotipo por ambiente para peso corporal y supervivencia a la cosecha en dos líneas genéticas de camarón blanco del Pacífico Penaeus (Litopenaeus) vannamei* [Estimation of genotype-by-environment interaction effects for body weight and survival at harvest in two genetic lines of Pacific white shrimp] [Doctoral dissertation, Universidad Nacional Autónoma de México]. UNAM Digital Repository. <https://repositorio.unam.mx/contenidos/3429271>
2. Cala-Moreno, N. (2013). *Evolución de la variabilidad genética para peso corporal a edad comercial en una población seleccionada de camarón blanco del Pacífico Penaeus (Litopenaeus) vannamei* [Evolution of genetic variability for body weight at commercial age in a selected population of Pacific white shrimp] [Master's thesis, Universidad Nacional Autónoma de México]. UNAM Digital Repository. <https://repositorio.unam.mx/contenidos/3479511>

### Peer-Reviewed Publications

1. Gómez-Rojas, C. J., Martínez-Marín, G., Toledo-Alvarado, H. O., & Cala-Moreno, N. (2026). Genetic parameters of larval weight and length in yellow mealworm under tropical conditions: heritability and implications for breeding programs. *animal*, 101784. <https://doi.org/10.1016/j.animal.2026.101784>
2. Cala-Moreno, N., Caicedo Blanco, Y. A., García Arévalo, J. L., & Toledo Alvarado, H. O. (2025). La cabra criolla santandereana: patrimonio genético y sostenibilidad productiva en el nororiente colombiano. *Spei Domus*, 21(1), 1–16. <https://doi.org/10.16925/2382-4247.2025.01.11>
3. Pulido Montes, M. A., Rincón Romero, K., Camargo Angel, A. L., Moreno Medina, D. F., Rojas Maestre, C. C., Serna Villamizar, K. J., & Cala Moreno, N. (2025). Palliative care nursing interventions to improve quality of life in oncological patients. *Salud, Ciencia y Tecnología*, 5, 1620. <https://sct.ageditor.ar/index.php/sct/article/view/1620>
4. Ortiz-Padilla, W., Campos-Montes, G. R., Cala-Moreno, N., & Caballero-Zamora, A. (2024). Cría y comercialización de insectos comestibles como fuente alternativa de proteína a nivel global. *Spei Domus*, 20(1), 1–22. <https://doi.org/10.16925/2382-4247.2024.01.02>
5. Soto, S. G. N., Villalobos, J. M. B., Moreno, N. C., Monforte, J. G. M., Galván, P. O., Arvisu, R. U., & Alvarado, H. O. T. (2023). Genome-wide association study for heat stress resistance in Brown Swiss cattle in Yucatan, Mexico. *Veterinaria México OA*, 10.
6. Cala Moreno, N., Hernández Rojas, I. K., Merchán Amaya, D. M., Miranda Fontecha, J. D., & Morales López, P. A. (2022). Influencia del COVID-19 en las dinámicas de exportación, producción y consumo de carne vacuna en Colombia y el mundo. *Spei Domus*, 18(1), 1–21. <https://doi.org/10.16925/2382-4247.2022.01.04>

7. Cala-Moreno, N., Campos-Montes, G., Caballero-Zamora, A., Berruecos-Villalobos, J., & Castillo-Juárez, H. (2021). Genotype-by-environment interaction in White shrimp associated with White Spot Disease. *Abanico Veterinario*, (11), 1–16. <https://dx.doi.org/10.21929/abavet2021.13>
8. Madalena, F. E., Toledo-Alvarado, H., & Cala-Moreno, N. (2019). Animals That Produce Dairy Foods: *Bos indicus* Breeds and *Bos indicus* × *Bos taurus* Crosses. *Reference Module in Food Science*, Elsevier. <https://doi.org/10.1016/B978-0-08-100596-5.00619-3>

### Conference Papers and Proceedings

1. Cala-Moreno, N., Montaldo, H. H., Ruíz-López, F. J., & Castillo-Juárez, H. (2015, November 25–27). Deterministic modeling of breeding programs that incorporate genomic selection for the Mexican Holstein population [Conference presentation abstract]. Reunión Nacional de Investigación Pecuaria, 1(1), 207–209. <https://doi.org/10.6084/m9.figshare.14357465.v1>
2. Cala-Moreno, N., Montaldo, H., Campos-Montes, G., & Castillo-Juárez, H. (2014, August 17–22). Evolution of genetic variability for growth-out survival rate in a selected population of Pacific white shrimp *Penaeus* (*Litopenaeus*) *vannamei* [Conference presentation abstract]. 10th World Congress of Genetics Applied to Livestock Production. <https://doi.org/10.6084/m9.figshare.14363810.v2>
3. Montaldo, H. H., Cala-Moreno, N., Ruíz-López, F. J., Campos-Montes, G. R., & Castillo-Juárez, H. (2013, August 9–12). Evolution of the genetic variability for body weight at harvest size in a selected population of Pacific White shrimp *Penaeus* (*Litopenaeus*) *vannamei* [Conference presentation abstract]. Aquaculture Europe — European Aquaculture Society Annual Meeting, Trondheim, Norway. <https://doi.org/10.6084/m9.figshare.14818812.v1>
4. Cala-Moreno, N., Montaldo, H. H., Castillo-Juárez, H., Campos-Montes, G. R., & Ruíz-López, F. J. (2012, September 10–13). Evolution of genetic variability for weight at commercial age in a selected population of Pacific white shrimp [*Penaeus* (*Litopenaeus*) *vannamei*] [Conference presentation abstract]. III Reunión Nacional de Innovación Acuícola y Pesquera, Querétaro, Mexico. <https://doi.org/10.6084/m9.figshare.14360069>
5. Cala-Moreno, N., Montaldo, H. H., Castillo-Juárez, H., Campos-Montes, G. R., & Ruíz-López, F. J. (2012, September). Evolution of genetic variability for weight at commercial age in a selected population of Pacific white shrimp [*Penaeus* (*Litopenaeus*) *vannamei*] [Conference abstract]. 2nd Annual Congress of the Department of Crop and Animal Production, Universidad Autónoma Metropolitana, Mexico City, Mexico. <https://doi.org/10.6084/m9.figshare.14818779.v1>
6. Tapias, J. A., Cala-Moreno, N., & Arcila Q., V. H. (2007, November 1–2). Development of an epidemiological map of zoonotic diseases in Santander, Colombia [Conference presentation abstract]. IX Encuentro Nacional y II Internacional de Investigadores de las Ciencias Pecuarias. *Revista Colombiana de Ciencias Pecuarias*, 20(4), 558–559. <https://doi.org/10.6084/m9.figshare.14363648.v1>

### Conference Posters

1. Cala-Moreno, N., Montaldo, H. H., Ruíz-López, F. J., Campos-Montes, G. R., & Castillo-Juárez, H. (2013, November 20–23). Evolution of the genetic variability for body weight at harvest size in a selected population of Pacific White shrimp *Penaeus* (*Litopenaeus*) *vannamei* [Poster session]. 3rd International Congress on Disease Ecology, Mérida, Yucatán, Mexico. <https://doi.org/10.6084/m9.figshare.14367092.v1>
2. Cala-Moreno, N., Montaldo, H. H., Castillo-Juárez, H., Campos-Montes, G. R., & Ruíz-López, F. J. (2012, September). Evolution of genetic variability for weight at commercial age in a selected population of Pacific white shrimp [*Penaeus* (*Litopenaeus*) *vannamei*] [Poster session]. 2nd Annual Congress of the Department of Crop and Animal Production, Universidad Autónoma Metropolitana, Mexico City, Mexico. <https://doi.org/10.6084/m9.figshare.14364392.v1>

### Invited Talks and Academic Presentations

1. Cala-Moreno, N. (2021, May 4). Estimation of genotype-by-environment interaction effects for body weight and survival at harvest in two genetic lines of Pacific white shrimp (*Penaeus* (*Litopenaeus*) *vannamei*) reared in ponds affected and unaffected by White Spot Disease [PhD dissertation defence]. Faculty of Veterinary Medicine and Animal Science, UNAM, Mexico City, Mexico. <https://doi.org/10.6084/m9.figshare.14819118.v1>
2. Cala-Moreno, N. (2018, January 11). Estimation of effects and genotype-by-environment interaction for traits of zootechnical interest in a population of Pacific white shrimp (*Penaeus* *vannamei*) in Mexico [PhD candidature presentation]. Faculty of Veterinary Medicine and Animal Science, UNAM, Mexico City, Mexico. <https://doi.org/10.6084/m9.figshare.14819355.v1>

3. Cala-Moreno, N. (2014, October 7). Evolution of genetic variability for growth and survival in a population of Pacific white shrimp [Conference presentation]. Institutional Seminars, Faculty of Veterinary Medicine and Animal Science, UNAM, Mexico City, Mexico. <https://doi.org/10.6084/m9.figshare.14367962.v1>
4. Cala-Moreno, N. (2013, June). Optimisation of breeding programmes using genomic information [PhD application presentation]. Faculty of Veterinary Medicine and Animal Science, UNAM, Mexico City, Mexico. <https://doi.org/10.6084/m9.figshare.14368019.v1>
5. Cala-Moreno, N. (2012, August). Evolution of genetic variability for weight at commercial age in a selected population of Pacific white shrimp *Penaeus (Litopenaeus) vannamei* [Conference presentation]. VII Congress of the Master's and Doctoral Programme in Animal Production and Health Sciences, UNAM, Mexico City, Mexico. <https://doi.org/10.6084/m9.figshare.14367911.v2>
6. Cala-Moreno, N. (2012, June). Genetic trend [Master's teaching presentation — Quantitative Genetics II]. Lecture on genetic trend, history and estimation. Faculty of Veterinary Medicine and Animal Science, UNAM, Mexico City, Mexico. <https://prezi.com/jvrwuwdree0t/tendencia-genetica/>

### Conferences and Seminars Attended

1. Machine Learning in Genomics: Tools, Resources, Clinical Applications and Ethics. 04/2021
2. 15th International Aquaculture Forum. 04/2021
3. 1st AGROGENÓMICA Meeting. 03/2021
4. II Meeting of the Young Researchers Seedbed. 06/2002
5. II Seminar on Advances in Small Animal Clinical Practice. 10/2001
6. First Green Agricultural Seminar. 05/2000

## RESEARCH GROUPS

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Animal Sciences Research Group (GRICA), Faculty of Veterinary Medicine and Animal Science, Universidad Cooperativa de Colombia — Category A1, Colciencias (Colombian Ministry of Science). Active collaboration with the Nursing Research Group of the Universidad Cooperativa de Colombia.

## RESEARCH PROJECTS

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1. **Estimation of genetic parameters for production traits in a genetic nucleus of *Tenebrio molitor* under tropical conditions.** Principal investigator: Cala-Moreno, N. Co-investigators: Caballero-Zamora, A.; Ortiz-Padilla, W. Mid-scale funded project, in collaboration with Universidad Autónoma Metropolitana — Xochimilco, Mexico City, Mexico.
2. **Effects of packaging and storage time on the nutritional, organoleptic and microbiological quality of red tilapia (*Oreochromis* spp.) in a Biofloc technology system.** Principal investigator: Cala-Delgado, D. Co-investigators: Cala-Moreno, N.; Hernández-Martínez, I. Y. Mid-scale funded project.
3. **Genome-wide association study for heat stress and gastrointestinal parasite tolerance in a native population of Santandereana Creole goat.** Principal investigator: Cala-Moreno, N. Co-investigators: Toledo-Alvarado, H. O.; Caicedo-Blanco, Y. A.; García-Arévalo, J. L. Mid-scale funded project, in collaboration with Universidad Nacional Autónoma de México, Mexico City, Mexico.
4. **Physiological, haematological and biochemical profile in native goats under gastrointestinal parasitism by *Trichostrongylus* spp. and *Haemonchus contortus*.** Principal investigator: Cala-Moreno, N. Co-investigators: Jaimes-Dueñez, J. E.; Toledo-Alvarado, H. O. Mid-scale project in preparation, in collaboration with Universidad Nacional Autónoma de México, Mexico City, Mexico.
5. **Genetic variability for production traits in a genetic nucleus of *Zophobas morio* (Coleoptera: Tenebrionidae).** Principal investigator: Cala-Moreno, N. Co-investigators: Cala-Delgado, D.; Toledo-Alvarado, H. O. Mid-scale project in preparation, in collaboration with Universidad Nacional Autónoma de México, Mexico City, Mexico.
6. **Genome-wide association study for heat stress and haemoparasite tolerance in a native population of the Chino Santandereano breed.** Principal investigator: Cala-Moreno, N. Mid-scale project in preparation.

## SUPERVISED THESES AND DISSERTATIONS

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Universidad Cooperativa de Colombia — Bucaramanga, unless otherwise stated.

1. *Clostridium butyricum* as a probiotic for the control of *Salmonella* spp. in laying hens. Master's thesis, completed, 2026.
2. Genetic parameters for larval weight and length in yellow mealworm (*Tenebrio molitor*) under tropical conditions. Master's thesis, completed, 2026.
3. Assessment of the impact of biological, environmental and genetic factors associated with pregnancy rate in in vitro embryo transfer programmes in cattle. Master's thesis, completed, 2025.
4. Association between gastrointestinal parasitism (*Haemonchus contortus* and *Trichostrongylus* spp.) and haematological parameters in the Santandereana Creole goat. Completed, 2026.
5. Health status modelling in the Santandereana Creole goat through the integration of parasite load, haematological parameters and clinical variables. Completed, 2026.
6. Genome-wide association for milk production and conformation traits in goats: a systematic review, 2010–2025. Completed, 2026.
7. Polystyrene biodegradation in the diet of *Tenebrio molitor*: a systematic review. Completed, 2026.
8. Practical manual for the management of small animal emergencies using ultrasonography at the Bivett veterinary centre. Completed, 2025.
9. Gastrointestinal parasitism (*Haemonchus contortus* and *Trichostrongylus* spp.) and physiological constants in the Santandereana Creole goat. Completed, 2025.
10. Effect of different feed substrates on larval weight in *Tenebrio molitor*: a systematic review with meta-analysis. Completed, 2025.
11. Oestrus synchronisation protocols in buffalo (*Bubalus bubalis*): a systematic literature review. Completed, 2022.
12. Phenotypic trend for body weight and its association with gastrointestinal parasitism in the Santandereana Creole goat. DVM dissertation, completed, 2024.
13. Weight, length and survival at the larval stage of *Tenebrio molitor* reared under two substrates and two photoperiods. DVM dissertation, completed, 2024.
14. Fitted lactation curve models for calcium in Holstein cows. DVM dissertation, completed, 2024.
15. Fitted lactation curve models for phosphorus in Holstein cows in Mexico. DVM dissertation, completed, 2024.
16. Physiological variables in the Santandereana Creole goat and their association with gastrointestinal parasitism by *Haemonchus contortus* and *Trichostrongylus*. DVM dissertation, completed, 2024.
17. Factors affecting growth and survival in Nile tilapia (*Oreochromis niloticus*): a systematic review with meta-analysis. DVM dissertation, completed, 2024.
18. Factors affecting the survival of *Tenebrio molitor*: a systematic review. DVM dissertation, completed, 2023.
19. Heritability for larval length using traditional methods in the mealworm (*Tenebrio molitor*). DVM dissertation, completed, 2023.
20. Effect of different substrates and stocking densities on larval weight in *Tenebrio molitor*: a systematic review with meta-analysis. DVM dissertation, completed, 2023.
21. Influence of COVID-19 on the dynamics of beef export, production and consumption in Colombia and worldwide: a monographic review. DVM dissertation, completed, 2022.
22. Genetic selection and improvement as a determinant of beef quality and productivity. DVM dissertation, completed, 2022.
23. Zootechnical parameters of the F1 cross between COBB 500 and *Gallus domesticus*: a case report. DVM dissertation, completed, 2022.

## PROFESSIONAL EXPERIENCE

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### Full-Time Research Professor

*Universidad Cooperativa de Colombia · Bucaramanga, Colombia · February 2022 – June 2026*

- Undergraduate and graduate teaching in genetics, quantitative genetics, biostatistics, experimental design, research methodology and academic writing.
- Supervision of undergraduate dissertations and master's theses in animal genetics, animal production, epidemiology and systematic reviews.
- Design, management and delivery of research projects in genetics, genomics, sustainable production and animal health.
- Production of peer-reviewed articles, outreach articles and other research outputs.
- Curriculum design and updating for undergraduate and graduate programmes.
- Methodological and statistical consulting for institutional research projects.
- Participation in quality assurance, self-assessment and institutional academic strengthening processes.

#### KEY ACHIEVEMENTS

- Supervision of more than 20 completed dissertations and theses.
- Principal investigator on funded projects in *Tenebrio molitor* and the Santandereana Creole goat.
- Publication of articles in national and international peer-reviewed journals.
- Contribution to the academic design of graduate programmes.
- Development of scientific, academic and knowledge-transfer outputs from institutional projects.

### Adjunct Lecturer

*Universidad Cooperativa de Colombia · Bucaramanga, Colombia · August 2021 – December 2022*

- Teaching in quantitative genetics, animal breeding, experimental design and academic writing.
- Planning, delivery and assessment of teaching and learning activities under the institutional educational model.
- Academic mentoring and guidance of undergraduate students.
- Design of activities aimed at critical thinking, research and problem solving.
- Integration of information and communication technologies into teaching processes.

#### KEY ACHIEVEMENTS

- Appointed to undergraduate and graduate programmes as a specialist lecturer in genetics and quantitative analysis.
- Supervision of academic work in animal genetics and research methodology.
- Methodological and statistical support for master's students.

### Research Projects Consultant

*Independent · January 2018 – Present*

- Methodological and statistical consulting for research projects in animal genetics, animal production and veterinary sciences.
- Experimental design and data analysis using mixed models, multivariate analysis and advanced statistical tools.
- Support in project design and in undergraduate, master's and doctoral theses.
- Support in scientific writing, results analysis and interpretation of complex biological information.
- Consultancy on genomic selection, GWAS, quantitative genetics and sustainable animal production.

#### KEY ACHIEVEMENTS

- Participation in international projects with researchers from Colombia and Mexico.
- Methodological supervision for undergraduate, master's and doctoral theses.
- Support for indexed scientific publications and funded projects.



## Genetic Improvement Advisor

Colombia · December 2015 – May 2017

- Genetic evaluation of registered cattle populations using statistical and genomic tools.
- Development of deterministic models to evaluate genomic selection strategies in Colombian Simmental cattle.
- Analysis and interpretation of pedigree, production and genomic information to support animal breeding programmes.
- Technical advice to organisations and producers involved in genetic selection programmes.

### KEY ACHIEVEMENTS

- Contribution to genomic selection implementation strategies for the Simmental breed in Colombia.
- Application of quantitative methodologies to optimise cattle breeding programmes.

## Specialised Sales Representative

Federico Osorio A. S.A. · Barranquilla, Colombia · May 2009 – July 2010

- Commercialisation of livestock inputs and technical training on the use of veterinary products.
- Management and training of the sales team.

## Rainbow Trout Production Instructor

Servicio Nacional de Aprendizaje (SENA) · Santander, Colombia · May 2008 – December 2008

- Training of rainbow trout producers in the province of Santa Bárbara, Santander.
- Development of a rainbow trout grow-out programme and management of public funds for its implementation.
- Monitoring and consultancy for emerging small-scale trout enterprises.

## Sales Manager

Laboratorios V.M. Ltda. · Bogotá, Colombia · January 2007 – October 2007

- Commercialisation of livestock inputs and management of veterinary product use in domestic animals.
- Management and training of the sales team; staff supervision.

## Veterinary Store Manager

Fondo Ganadero de Santander (FOGASA) · San Alberto, Cesar, Colombia · June 2006 – December 2006

- Veterinary care for livestock operations in the area.
- Commercialisation and administration of livestock inputs; purchasing and inventory management.
- Management of sales and administrative staff, and of the branch's financial resources.

## RELATED ACADEMIC EXPERIENCE

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### Teaching Assistant

Universidad Nacional Autónoma de México · Mexico City, Mexico · February 2018 – December 2019

- Development of new lecture content and methodological teaching proposals.
- Support to faculty in marking reports and assignments.
- Teaching and tutoring undergraduate students in Animal Breeding and Statistics.

### Teaching Assistant

UNAM and Universidad Autónoma Metropolitana — Xochimilco · Mexico City, Mexico · February 2016 – December 2017

- Support to faculty in marking reports and assignments.
- Teaching and tutoring undergraduate students in Animal Breeding and Statistics.
- Teaching and tutoring master's and doctoral students in Quantitative Genetics and Statistics.

### Teaching Assistant

Universidad Nacional Autónoma de México · Mexico City, Mexico · August 2013 – June 2015

- Teaching and tutoring undergraduate and master's students in Animal Breeding, Quantitative Genetics and Statistics.
- Training undergraduate students in the data analysis laboratory using SPSS and R.

## Veterinary Intern, Poultry Processing Plant

Pollosan S.A. · Bucaramanga, Colombia · June 2004 – November 2004

- Supervision of staff and broiler transport.
- Quality control and veterinary supervision in the bird reception area of the processing plant.

## Teaching Aide

Instituto de Enseñanza Media Diversificada (INEM) · Bucaramanga, Santander, Colombia · June 2003 – December 2003

- Support to technical secondary students in livestock subjects.
- Support to faculty in marking reports and assignments.

## PEER REVIEW ACTIVITY

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1. Institutional research project reviewer — Universidad Cooperativa de Colombia. 11/2011
2. Manuscript reviewer — *Turkish Journal of Fisheries and Aquatic Sciences*. 12/2020
3. Manuscript reviewer — *Spei Domus*. 12/2023

## SOFTWARE AND TECHNICAL SKILLS

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ASReml — Advanced

JMP — Advanced

R — Advanced

MATLAB — Intermediate

SPSS — Intermediate

Python — Basic

## LANGUAGES

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Spanish — Native

English — Intermediate/Advanced

## INTERESTS

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Swimming, running, cycling, reading, playing the guitar, painting, photography and photo editing.

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